



# Introduction to Database Systems

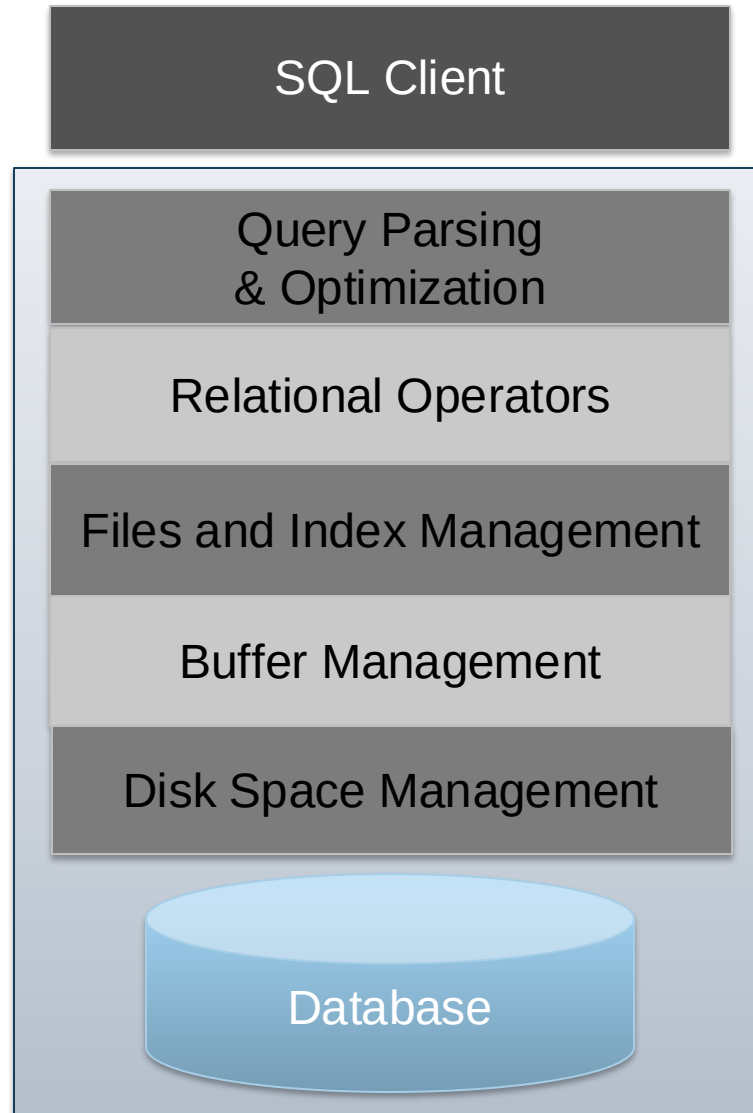
2024-Fall



## 4.4 Transactions Management

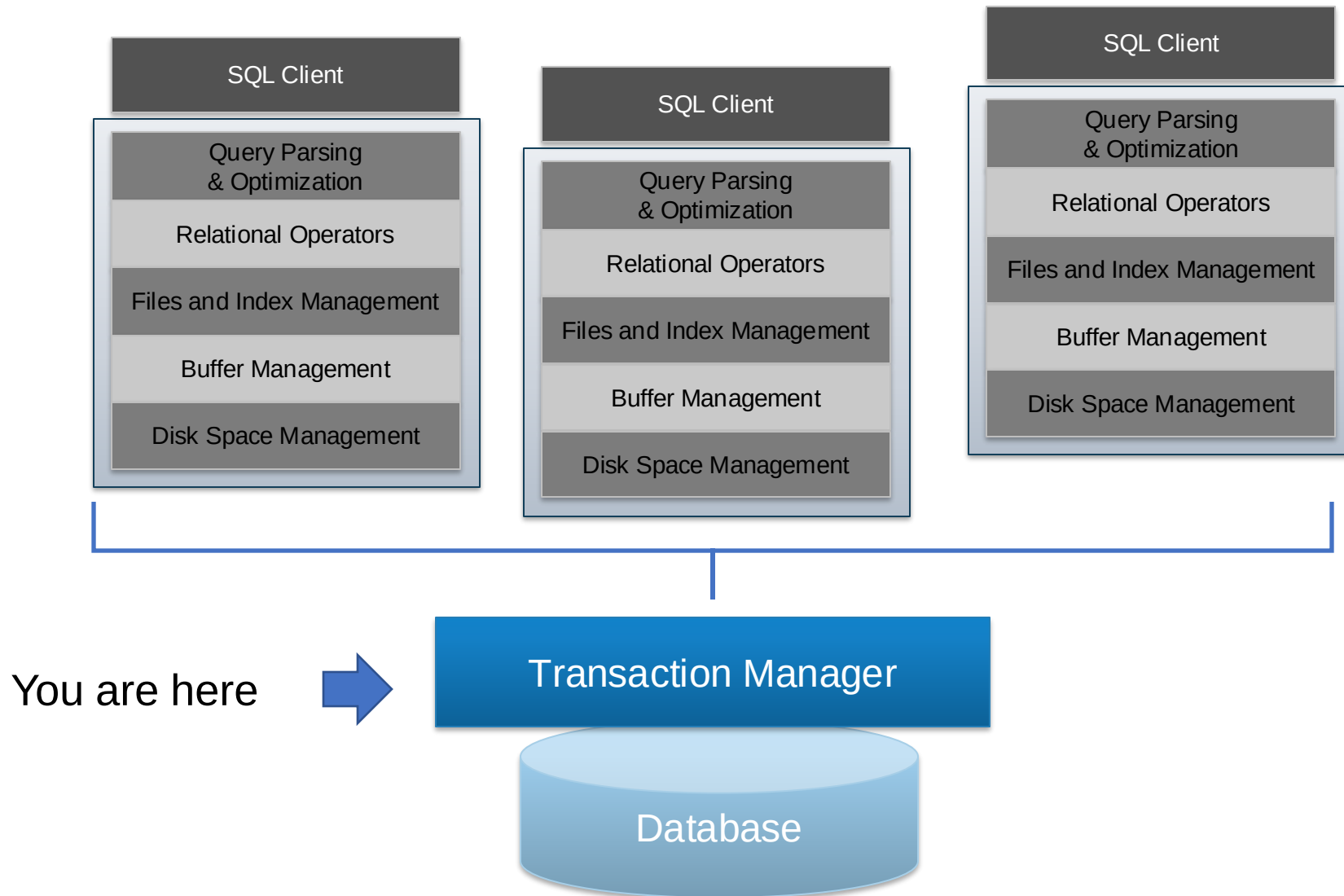


# Architecture of a DBMS



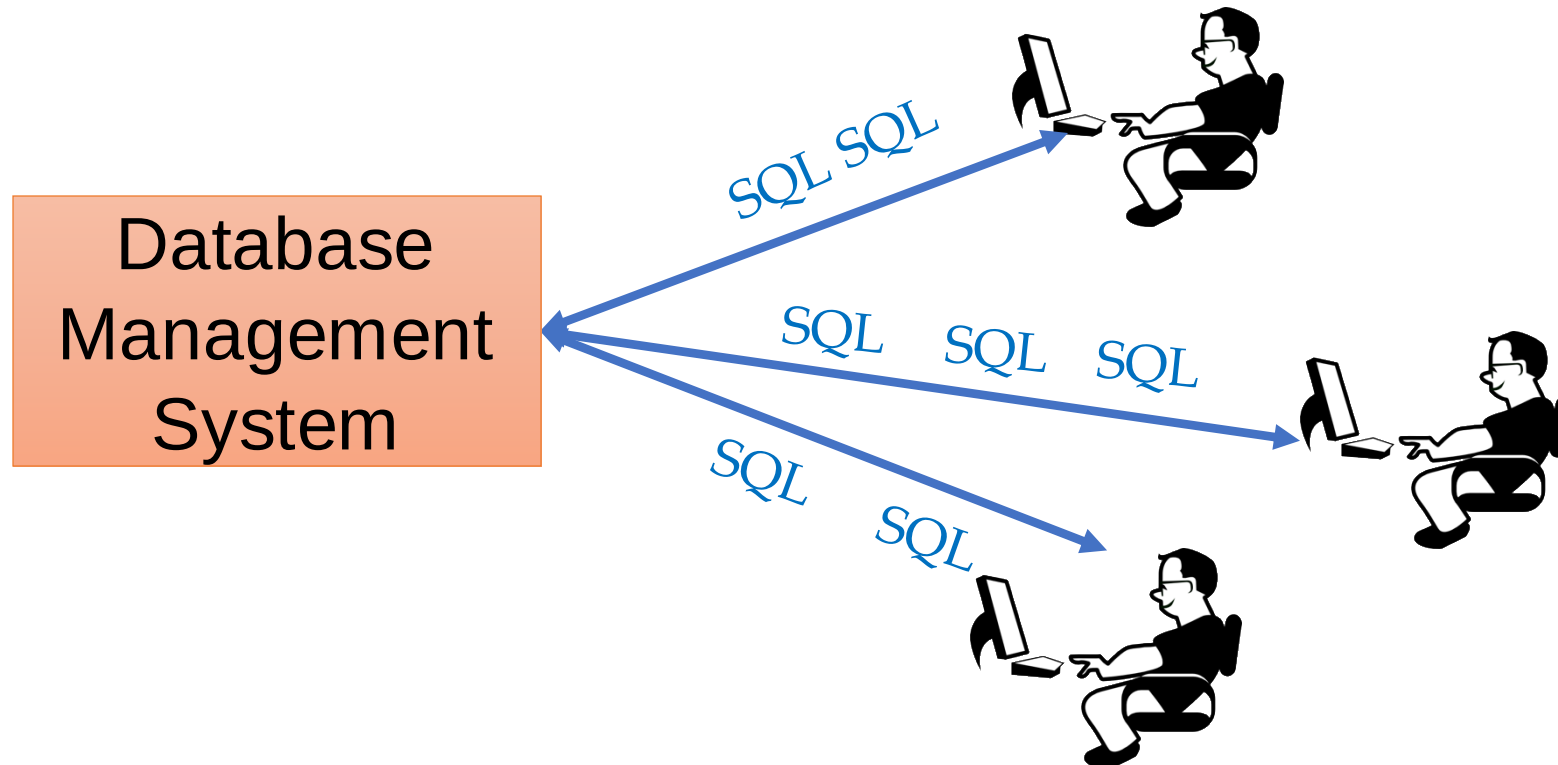


# Architecture of a DBMS, Part 2



# Applications Want Something from the DBMS

- Queries and updates of course: what you learned so far!
- Real applications are composed of many statements being generated by user behaviors
- Many users work with the application at the same time





# Transactions



# What is a Transaction?

- A *sequence of multiple actions* to be executed as an ***atomic unit***
- Application View (SQL View):
  - Begin transaction
  - Sequence of SQL statements
  - End transaction
- Examples
  - Transfer money between accounts
  - Book a flight, a hotel and a car together on Expedia



## 4.4 Transactions Management

- Recovery
- Structures to support recovery
- Update transactions
- Message
- Transaction failure
- Concurrency control
- Lock protocol





## 4.4.1 Recovery



# Recovery

## Introduction

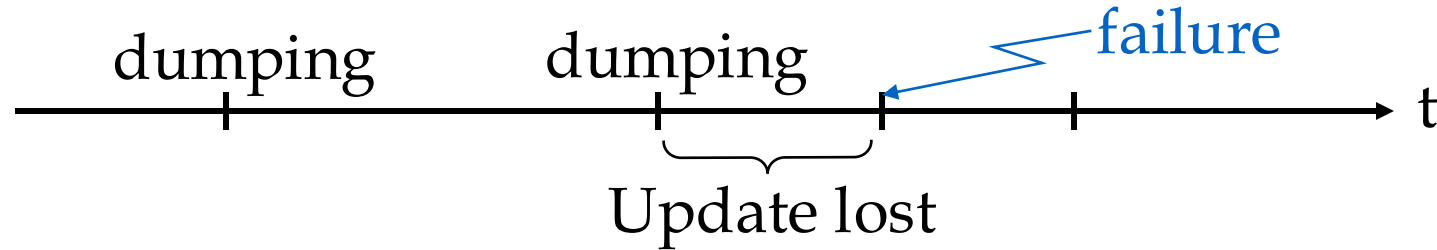
The main roles of recovery mechanism in DBMS are:

- (1) Reducing the likelihood of failures (prevention)
- (2) Recover from failures (solving)

**Restore DB to a consistent state after some failures.**

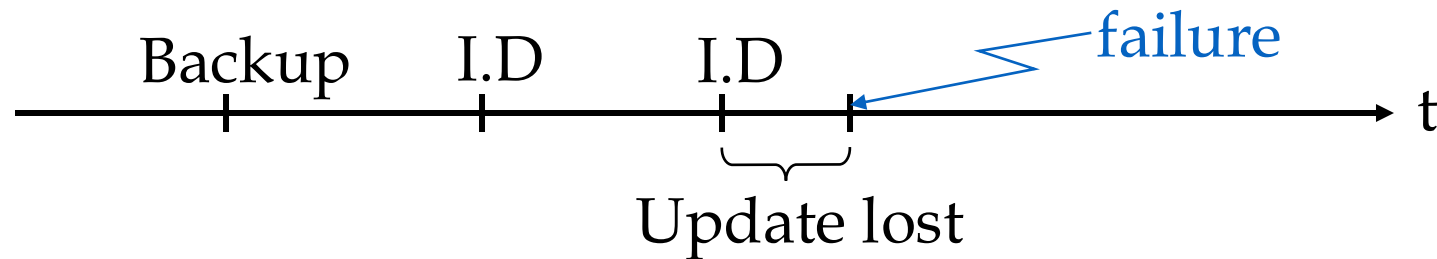
- Redundancy is necessary.
- Should inspect **all possible** failures.
- General method:
  - backup copies
  - backup copies and transaction logs
  - multiple copies

# 1) Backup——Periodical dumping



- Variation : Backup + Incremental dumping (增量转储)

I.D --- updated parts of DB



This method is easy to be implemented and the overhead is low, but the update maybe lost after failure occurring.

So it is often used in file system or small DBMS.

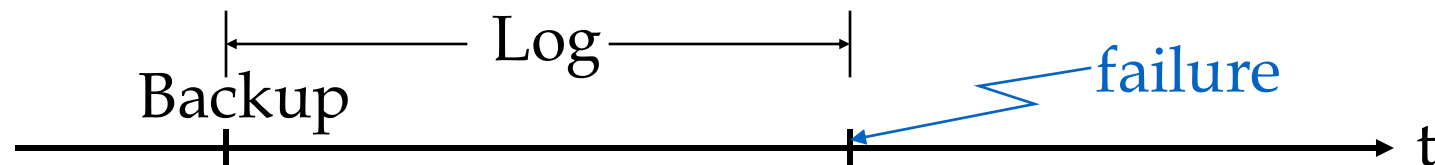


## 2) Backup + Log

Log : record of **all** changes on DB since the last backup copy was made.

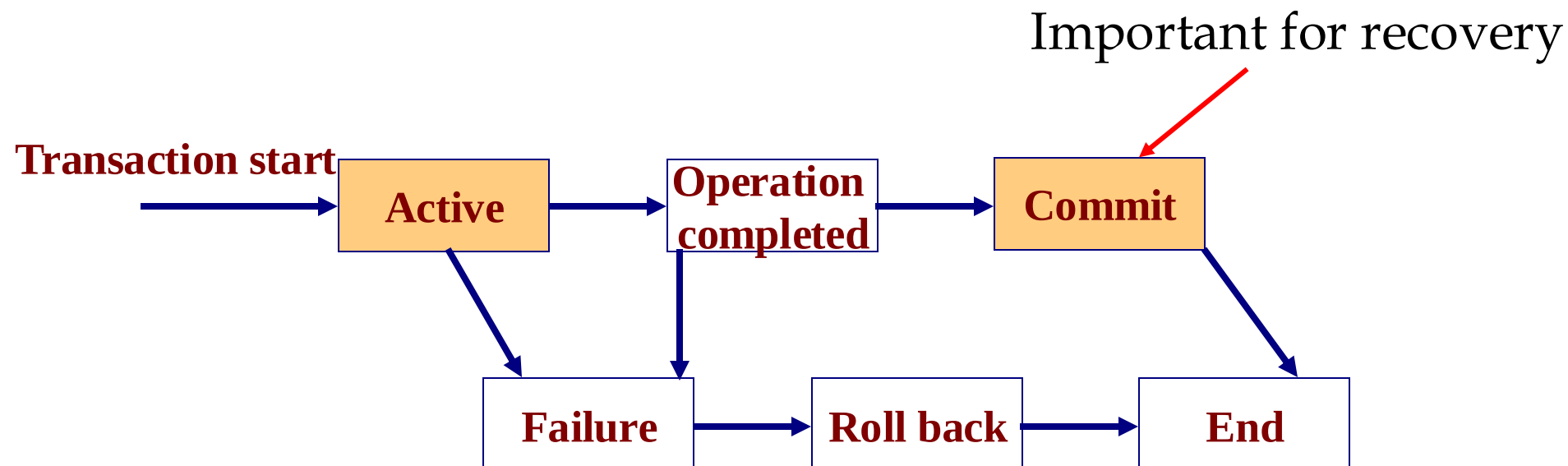
Change:  $\left\{ \begin{array}{l} \text{Old value (Before Image --- B.I)} \\ \text{New value (After Image --- A.I)} \end{array} \right\}$  Recorded into Log

|     |              |      |      |
|-----|--------------|------|------|
| For | update op. : | B.I  | A.I  |
|     | insert op. : | ---- | A.I  |
|     | delete op. : | B.I  | ---- |



# Transaction Status

- COMMIT/NOT COMMIT





# While recovering:

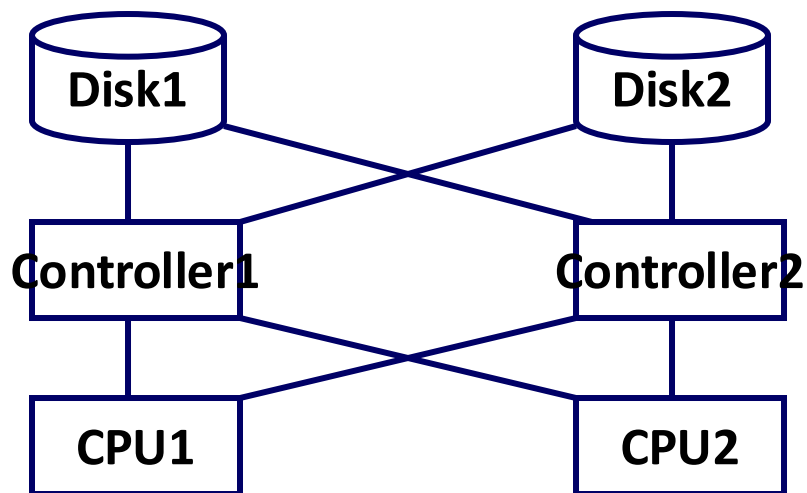
- Some transactions maybe half done, should undo them with B.I recorded in Log (backward recovery).
- Some transactions have finished but the results have not been written into DB in time, should redo them with A.I recorded in Log (forward recovery). (finish writing into DB)

It is possible to recover DB to the **most recent** consistent state with Log.



# Multiple copies

- If a system has multiple database copies and these copies have independent failure modes, they can be used as backups for each other for recovery purposes.
- This technique is commonly applied in distributed database systems.
- In recent years, due to the decrease in hardware prices, mirrored disks technology has also been adopted.



**mirrored disks**

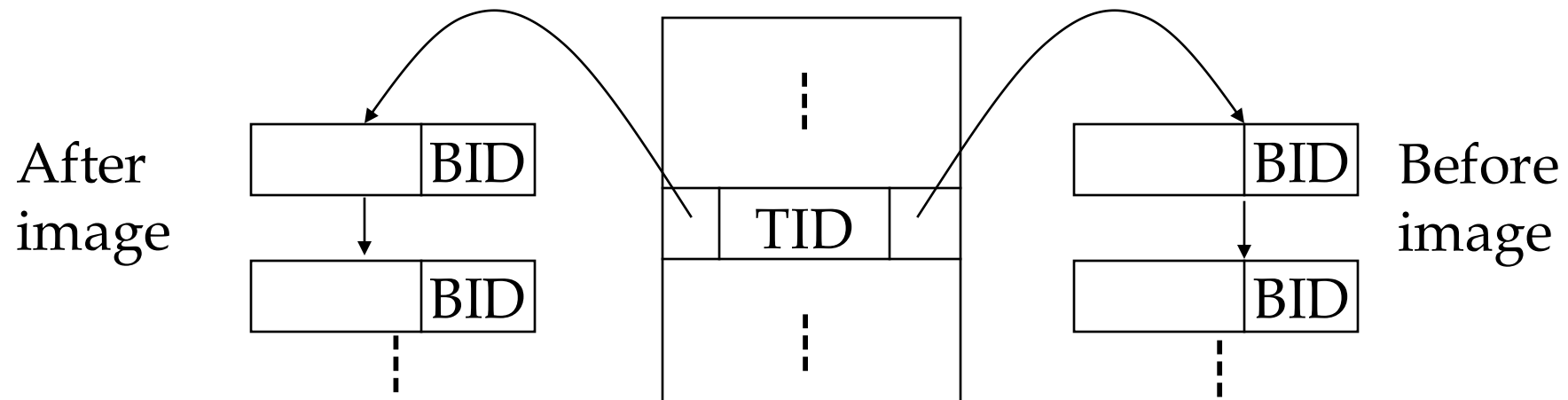
- When writing data, the same content is written to both disks. If the data on one disk is lost, it can be recovered using the data from the other disk. (The probability of both disks failing simultaneously can be assumed to be zero!)





## 4.4.2 Some Structures to Support Recovery (backup+log)

- Recovery information (such as Log) should be stored in **nonvolatile** storage.
- Log files generally should not be stored on the same disk as the database to prevent both from being lost.
- The following information need to be stored in order to support recovery:
  - 1) Commit list : list of TID which have been committed. **committed transaction list---CTL**
  - 2) Active list : list of TID which is in progress. **active transaction list---ATL**



When committing, first add the TID of the transaction to be committed to the CTL, then remove the corresponding TID from the ATL. Otherwise, if a failure occurs, the state of the transaction will be lost!



# Before image and After image

- B.I: It can be regarded as a heap file. Each physical block has a block identifier (BID), which consists of a *TID, relation name, and a logical block number*. The logical block number is unique within the relation. If a transaction needs to be rolled back, all the before-image blocks of that transaction can be found in the before-image file.
- A.I: The structure is similar to the before-image file, but it records after-images. During recovery, the after-images can be written back according to the transaction order in the committed transaction table and the logical block numbers. This is equivalent to redoing each transaction in the order they were committed.
- Idempotent (幂等性)
  - $\text{undo}(\text{undo}(\text{undo} \dots \text{undo}(x))) = \text{undo}(x)$
  - $\text{redo}(\text{redo}(\text{redo} \dots \text{redo}(x))) = \text{redo}(x)$



# Backup+log

- After taking a backup, the previous logs lose their value. For recovery purposes, only the logs after the most recent backup need to be retained. However, the logs may still be large. The following measures can be used to reduce the size of the logs.
  - Do not retain the before-images of committed transactions.
  - Selectively retain after-images.
  - Consolidate after-images (for a physical block with a given logical block number, if updated multiple times, only the most recent after-image needs to be retained).





## 4.4.3 Commit Rule and Log Ahead Rule

### 1) *Commit Rule* (提交规则)

A.I must be written to nonvolatile storage before commit of the transaction.

### 2) *Log Ahead Rule* (先记后写规则)

If A.I is written to DB before commit then B.I must first written to log.

### 3) *Recovery strategies*



# Three kinds of update strategy-1

***a) A.I → DB before commit***

TID → active list

B.I → Log (Log Ahead Rule)

A.I → DB, Log

⋮

TID → commit list

delete TID from active list

# The recovery after failure in this situation

- Check two lists for every TID while restarting after failure:

| Commit list | Active list |                                   |
|-------------|-------------|-----------------------------------|
|             | ✓           | Undo, delete TID from active list |
| ✓           | ✓           | delete TID from active list       |
| ✓           |             | nothing to do                     |



# Three kinds of update strategy-2

***b) A.I → DB    after commit***

TID → active list

A.I → Log                      (Commit Rule)

⋮

TID → commit list

A.I → DB

delete TID from active list





# The recovery after failure in this situation

- Check two lists for every TID while restarting after failure:

| Commit list | Active list |                                   |
|-------------|-------------|-----------------------------------|
|             | ✓           | delete TID from active list       |
| ✓           | ✓           | redo, delete TID from active list |
| ✓           |             | nothing to do                     |



# Three kinds of update strategy-3

## *c) A.I $\rightarrow$ DB concurrently with commit*

TID  $\rightarrow$  active list

A.I, B.I  $\rightarrow$  Log (Two Rules)

A.I  $\rightarrow$  DB (partially done)

⋮

TID  $\rightarrow$  commit list

A.I  $\rightarrow$  DB (completed)

delete TID from active list

# The recovery after failure in this situation

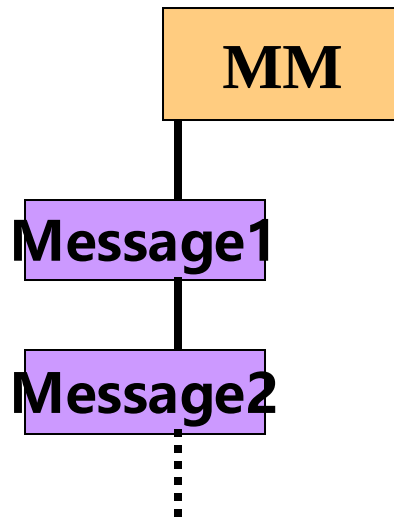
- Check two lists for every TID while restarting after failure:

| Commit list | Active list |                                   |
|-------------|-------------|-----------------------------------|
|             | ✓           | Undo, delete TID from active list |
| ✓           | ✓           | redo, delete TID from active list |
| ✓           |             | nothing to do                     |



## 4.4.4 Message Management

- A transaction often needs to send impactful messages to users, which are not the kind of messages that require user input for direction. The task of message sending is delegated to a Message Manager (MM) subsystem.
- During the execution of a transaction, the messages that need to be sent are passed to the MM, which establishes a message queue for each transaction.



# Message Management

- The MM allows for the addition and deletion of messages entrusted for sending by the transaction until the normal conclusion of the transaction; once the transaction ends, the MM stores the messages in non-volatile memory.
- When a transaction ends normally (including commit and rollback), it notifies the MM to send the messages; if the transaction is aborted due to a failure, the MM will discard the messages for that transaction.
- To ensure reliable delivery of messages to the concerned users, the MM employs a "send-acknowledge" method of transmission.





## 4.4.5 Failure Classification

- **Transaction failure :**
  - **Logical errors:** transaction cannot complete due to some internal error condition
  - **System errors:** the database system must terminate an active transaction due to an error condition (e.g., deadlock)
- **System crash:** a power failure or other hardware or software failure causes the system to crash.
  - **Fail-stop assumption:** non-volatile storage contents are assumed to not be corrupted by system crash
    - Database systems have numerous integrity checks to prevent corruption of disk data
- **Disk failure:** a head crash or similar disk failure destroys all or part of disk storage
  - Destruction is assumed to be detectable: disk drives use checksums to detect failures



# Recovering from Transaction Failure

- When recovering after transaction failure:
  - MM discards the message queue of the transaction;
  - Perform an undo operation if necessary;
  - Remove the transaction's TID from the ATL and release the resources occupied by the transaction.



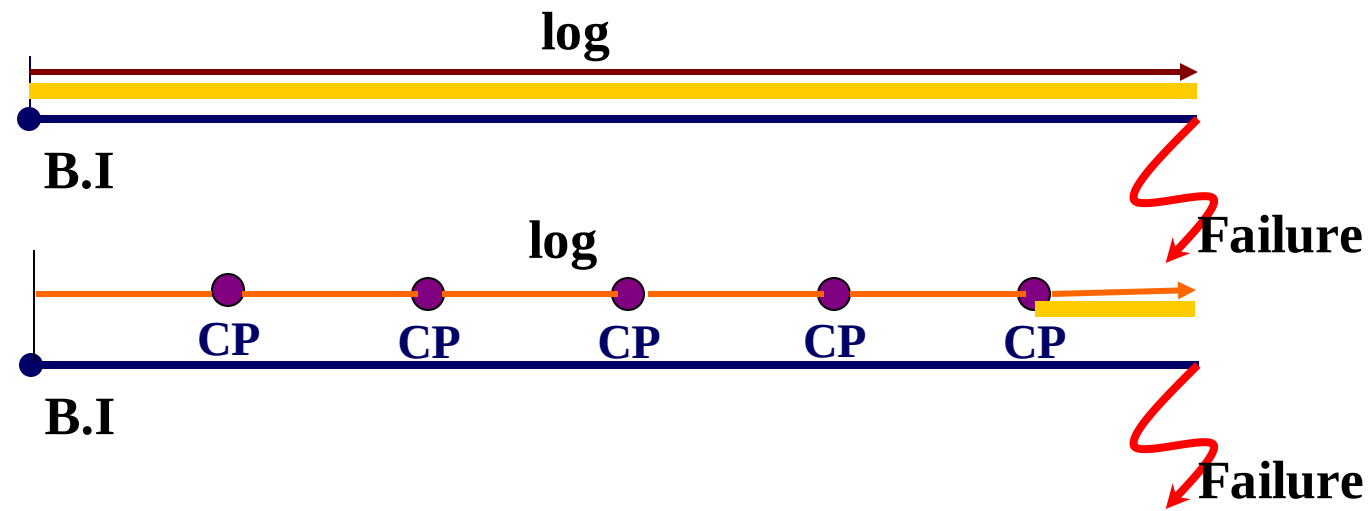


# Recovering from System Failure

- Restart the OS and DBMS;
- Restore the database to a consistent state (perform undo operations for uncommitted transactions and redo operations for committed transactions).
- Only when the database is restored to a consistent state are users allowed to access it.

# Checkpoints

- Redoing/undoing all transactions recorded in the log can be very slow
  - Processing the entire log is time-consuming if the system has run for a long time
  - We might unnecessarily redo transactions which have already output their updates to the database.
- Streamline recovery procedure by periodically performing **checkpointing**.
  - At a checkpoint, the DBMS forcefully writes the after-images of all committed transactions.





# Checkpoints (Cont.)

- The process of taking a checkpoint generally follows these steps:
  - *Pause* the execution of transactions;
  - Write the after-images of transactions committed since the last checkpoint;
  - Record the checkpoint in the log's control table (CTL);
  - Resume the execution of transactions.
- Taking a checkpoint significantly impacts the normal operation of the system, and only when a system failure occurs does it provide the benefit of reducing the amount of redo work.



# Recovering from Disk Failure

- Repair the system and update the disk if necessary;
- If the system (OS and DBMS) crashes, restart the system;
- Load the most recent backup copy;
- Redo all transactions committed after the most recent backup using the after-images from the log.



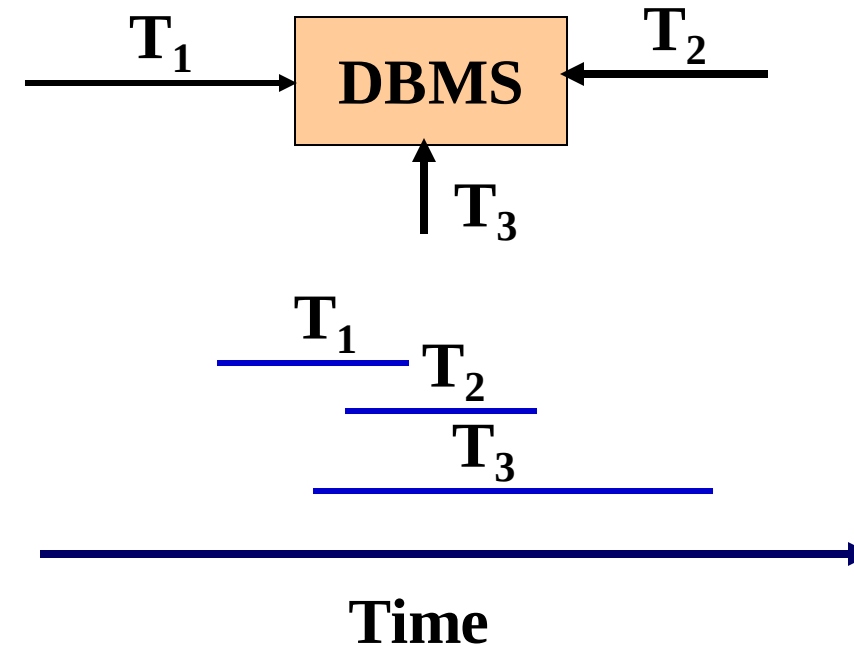
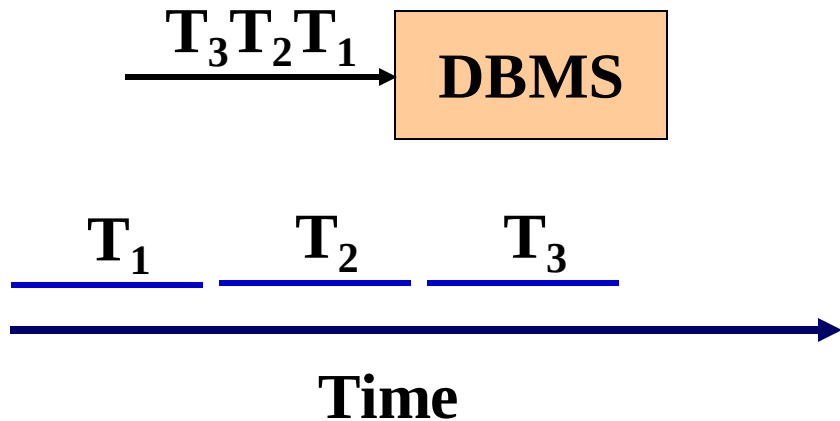
## 4.4.6 Concurrency Control



# Concurrency Control: Providing Isolation

- **Naïve approach - serial execution**
  - One transaction runs at a time
  - Safe but slow
- **Execution must be interleaved for better performance**
- With concurrent executions, how does one **define** and **ensure** correctness?

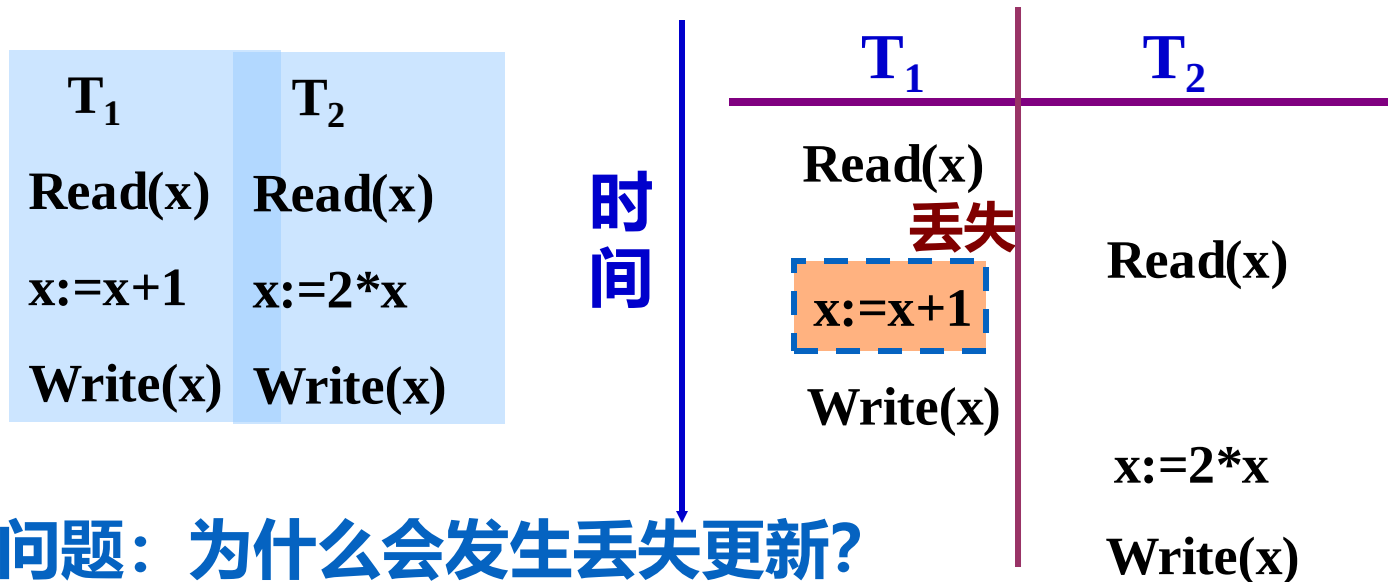
# Serial access vs. Concurrent access



Improve system resource utilization.  
Enhance the response time of short transactions.

# Issues of concurrency

- Lost update

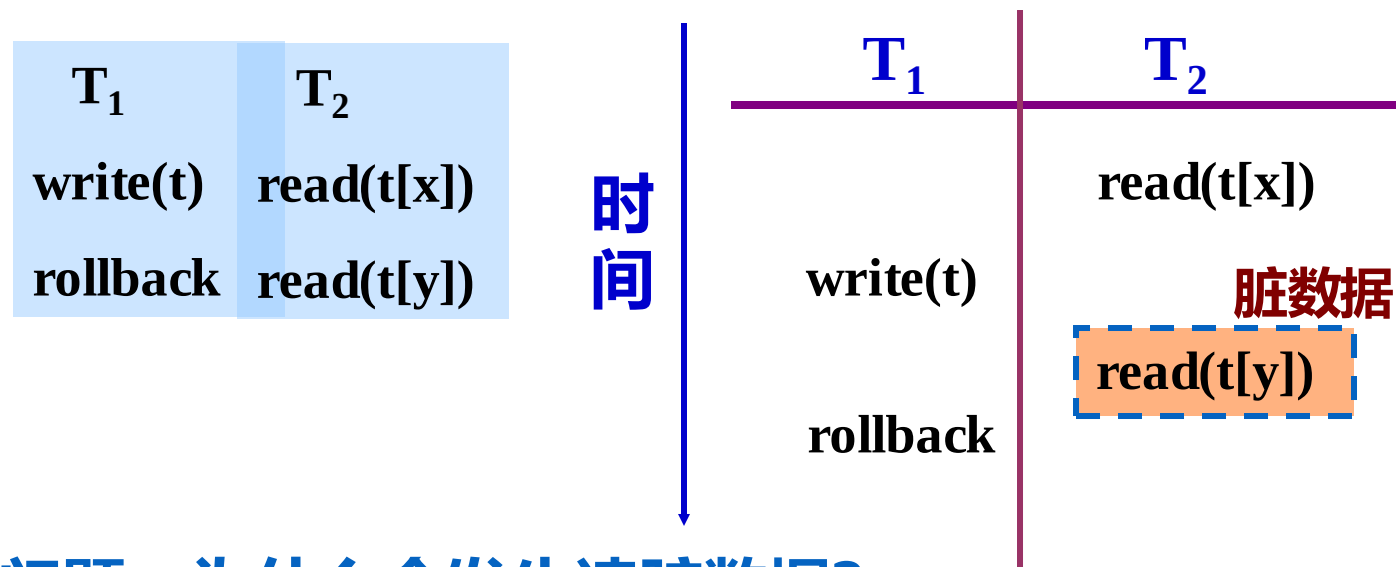


由两个事务对同一数据并发写入引起，称为“写-写冲突” (write-write conflict)



# Issues of concurrency

- Dirty read

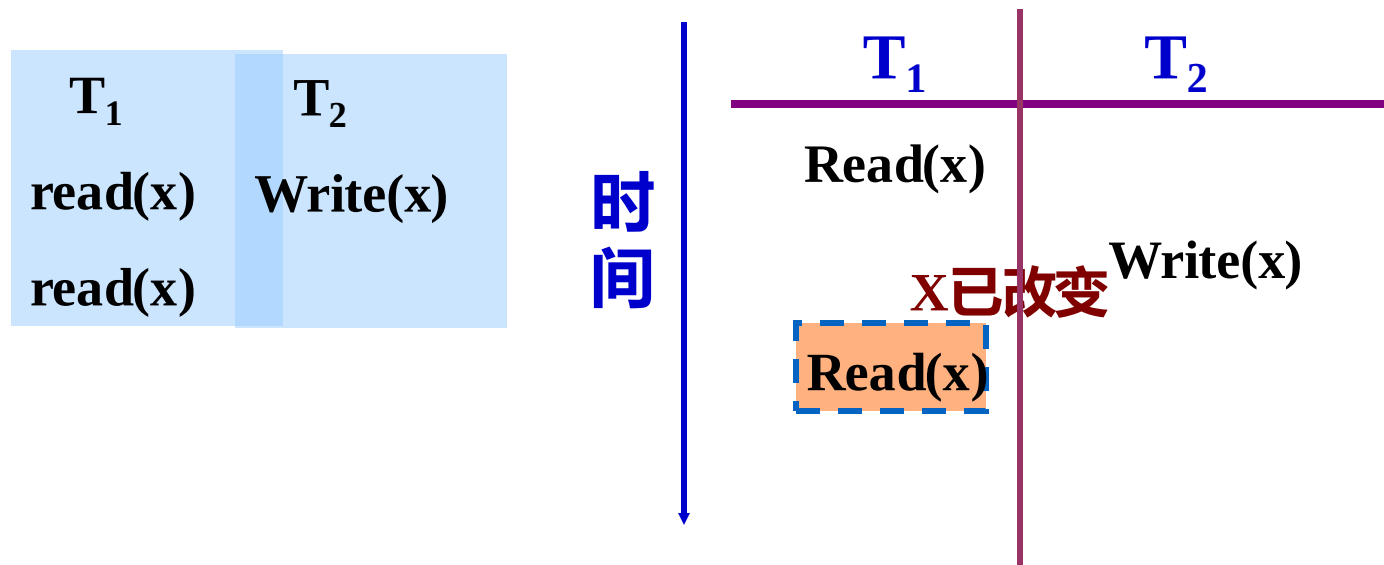


问题：为什么会发生读脏数据？

由于一个事务读取另一个更新事务尚未提交的数据引起，称为“读-写冲突” (read-write conflict)

# Issues of concurrency

- unrepeatable read



**问题：为什么会发生读值不可重现？**

由两个事务对同一数据并发读写引起，  
问题出在“写”操作上



# Conflicts

- Write-write
  - Write-write conflicts should be avoided in all cases
- Read-write
  - read-write conflicts should generally be avoided but can be tolerated in certain application scenarios.



# Transaction Schedules

| T1       | T2       |
|----------|----------|
| begin    |          |
| read(A)  |          |
| write(A) |          |
| read(B)  |          |
| write(B) |          |
| commit   |          |
|          | begin    |
|          | read(A)  |
|          | write(A) |
|          | read(B)  |
|          | write(B) |
|          | commit   |

A **schedule** is a sequence of actions on data from one or more transactions.

Actions: Begin, Read, Write, Commit and Abort.

$R_1(A) W_1(A) R_1(B) W_1(B) R_2(A) W_2(A) R_2(B) W_2(B)$

By convention we only include committed transactions, and omit Begin and Commit.



# Serial Equivalence

- We need a “touchstone” concept for correct behavior

- **Definition: Serial schedule**

- Each transaction runs from start to finish without any intervening actions from other transactions

- **Definition: 2 schedules are **equivalent** if they:**

- involve the same transactions
- each individual transaction’s actions are ordered the same
- both schedules leave the DB in the same final state

| T1    | T2    |
|-------|-------|
| read  |       |
| write |       |
| write |       |
| write |       |
| write |       |
| write |       |
|       | read  |
|       | write |
|       | write |
|       | write |
|       | write |
|       | write |

| T1    | T2    |
|-------|-------|
| read  |       |
| write |       |
| write |       |
| write |       |
| write |       |
| write |       |
|       | read  |
|       | write |
|       | write |
|       | write |
|       | write |
|       | write |

| T1    | T2    |
|-------|-------|
| read  |       |
| write |       |
| write |       |
| write |       |
| write |       |
| write |       |
|       | read  |
|       | write |
|       | write |
|       | write |
|       | write |
|       | write |



**View equivalence** (目标等价)

# Serializability

- **Definition:** Schedule S is **serializable** if:
  - S is equivalent to some serial schedule

[illegible]

**=?**

[illegible]

==?

| 例                                         | 例                                         |
|-------------------------------------------|-------------------------------------------|
|                                           | $\overline{f}fT \nearrow$                 |
|                                           | $\overline{f}fT\overline{f}/O_{10}$       |
|                                           | $\overline{f}fTfT\overline{f}/O_{10}$     |
|                                           | $\overline{f}fT\overline{f}\Delta_{10}$   |
|                                           | $\overline{f}fTfT\Delta_{10}$             |
|                                           | $\overline{f}fT\overline{f}T\overline{f}$ |
| $\overline{f}fT \nearrow$                 |                                           |
| $\overline{f}fT\overline{f}/O_{10}$       |                                           |
| $\overline{f}fTfT\overline{f}/O_{10}$     |                                           |
| $\overline{f}fT\overline{f}\Delta_{10}$   |                                           |
| $\overline{f}fTfT\Delta_{10}$             |                                           |
| $\overline{f}fT\overline{f}T\overline{f}$ |                                           |





# Schedule 1

| T1: Transfer \$100<br>from A to B | T2: Add 10%<br>interest to A & B |
|-----------------------------------|----------------------------------|
| begin                             |                                  |
| read(A)                           |                                  |
| A = A - 100                       |                                  |
| write(A)                          |                                  |
| read(B)                           |                                  |
| B = B + 100                       |                                  |
| write(B)                          |                                  |
| commit                            |                                  |
|                                   | begin                            |
|                                   | read(A)                          |
|                                   | A = A * 1.1                      |
|                                   | write(A)                         |
|                                   | read(B)                          |
|                                   | B = B * 1.1                      |
|                                   | write(B)                         |
|                                   | commit                           |

- Let T1 transfer \$100 from A to B
- Let T2 add 10% interest to A & B
- Serial schedule in which T1 is followed by T2
  - Final outcome:
    - $A := 1.1 * (A - 100)$
    - $B := 1.1 * (B + 100)$



# Schedule 2

| T1: Transfer \$100<br>from A to B | T2: Add 10%<br>interest to A & B |
|-----------------------------------|----------------------------------|
|                                   | begin                            |
|                                   | read(A)                          |
|                                   | $A = A * 1.1$                    |
|                                   | write(A)                         |
|                                   | read(B)                          |
|                                   | $B = B * 1.1$                    |
|                                   | write(B)                         |
|                                   | commit                           |
| begin                             |                                  |
| read(A)                           |                                  |
| $A = A - 100$                     |                                  |
| write(A)                          |                                  |
| read(B)                           |                                  |
| $B = B + 100$                     |                                  |
| write(B)                          |                                  |
| commit                            |                                  |

- Serial schedule in which T2 is followed by T1
  - Final outcome:
    - $A := (1.1 * A) - 100$
    - $B := (1.1 * B) + 100$
  - Different!
    - But still understandable





# Schedule 3

| T1: Transfer \$100<br>from A to B | T2: Add 10%<br>interest to A & B |
|-----------------------------------|----------------------------------|
| begin                             |                                  |
| read(A)                           |                                  |
| $A = A - 100$                     |                                  |
| write(A)                          |                                  |
|                                   | begin                            |
|                                   | read(A)                          |
|                                   | $A = A * 1.1$                    |
|                                   | write(A)                         |
| read(B)                           |                                  |
| $B = B + 100$                     |                                  |
| write(B)                          |                                  |
| commit                            |                                  |
|                                   | read(B)                          |
|                                   | $B = B * 1.1$                    |
|                                   | write(B)                         |
|                                   | commit                           |

- Schedule in which actions of T1 and T2 are interleaved.
- This is not a serial schedule
- But it is equivalent to schedule 1
  - $A := (A-100)*1.1$
  - $B := (B+100)*1.1$
- Hence **serializable**!



# Conflicting Operations

- Tricky to check property **“leaves the DB in the same final state”**
- **Use notion of “conflicting” operations (read/write)**
- **Definition: Two operations conflict if they:**
  - Are by different transactions,
  - Are on the same object,
  - At least one of them is a write.
- The order of non-conflicting operations has no effect on the final state of the database!
  - Focus our attention on the order of conflicting operations



# Conflict Serializable Schedules

- **Definition:** Two schedules are *conflict equivalent* (冲突等价) if:
  - They involve the same actions of the same transactions, and
  - Every pair of conflicting actions is ordered the same way
- **Definition:** Schedule *S* is *conflict serializable* if:
  - *S* is conflict equivalent to some serial schedule
  - Implies *S* is also Serializable
- **Note:** If two schedules are conflict equivalent, they must be result equivalent; however, the reverse is not necessarily true!



# Conflict Serializability - Intuition

- A schedule **S** is **conflict serializable** if
  - You are able to transform S into a serial schedule by swapping **consecutive non-conflicting** operations of different transactions
- *Example*

|           |           |
|-----------|-----------|
| R(A) W(A) | R(B) W(B) |
| R(A) W(A) | R(B) W(B) |



# Conflict Dependency Graph

- **Dependency Graph:**

- One node per Xact
- Edge from  $T_i$  to  $T_j$  if:
  - An operation  $O_i$  of  $T_i$  conflicts with an operation  $O_j$  of  $T_j$  **and**
  - $O_i$  appears earlier in the schedule than  $O_j$



- **Theorem: Schedule is conflict serializable if and only if its dependency graph is acyclic.**

Proof Sketch: Conflicting operations prevent us from “swapping” operations into a serial schedule



# Example

- A schedule that is not conflict serializable

T1: R(A), W(A)

T1

T2

*Dependency graph*



## Example, pt 2

- A schedule that is not conflict serializable

|     |             |
|-----|-------------|
| T1: | R(A), W(A), |
| T2: | R(A)        |



*Dependency graph*



## Example, pt 3

- A schedule that is not conflict serializable

T1: R(A), W(A),

T2: R(A), W(A), R(B), W(B)



*Dependency graph*

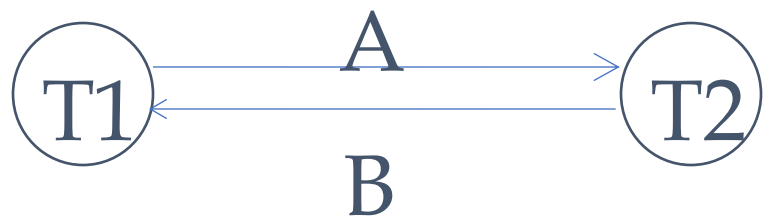




## Example, pt 4

- A schedule that is not conflict serializable

|     |                        |      |
|-----|------------------------|------|
| T1: | R(A), W(A),            | R(B) |
| T2: | R(A), W(A), R(B), W(B) |      |

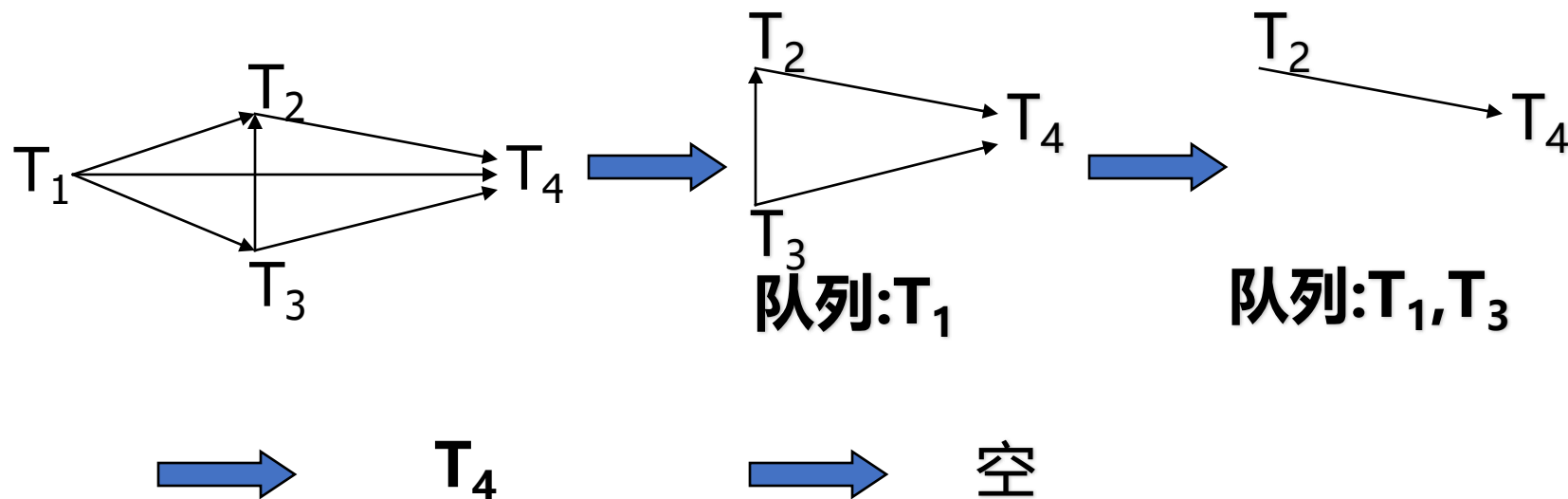


*Dependency graph*



# Example

$S = W_3(y)R_1(x)R_2(y)W_3(x)W_2(x)W_3(z)R_4(z)W_4(x)$



队列:  $T_1, T_3, T_2$

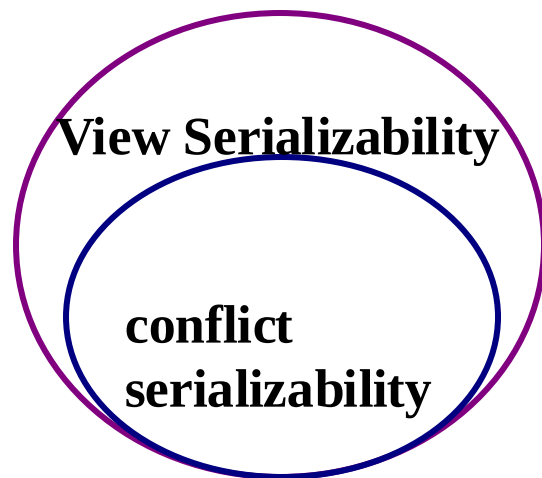
等价串行序列:  $T_1 \rightarrow T_3 \rightarrow T_2 \rightarrow T_4$

并发控制的任务就是对并发执行的事务加以控制，使之按某种可串行化的调度序列来执行。



# Notes on Serializability Definitions

- **View Serializability allows (a few) more schedules than conflict serializability**
  - But V.S. is difficult to enforce efficiently.
- **Neither definition allows all schedules that are actually serializable.**
  - Because they don't understand the meanings of the operations or the data
- **Conflict Serializability is what gets used, because it can be enforced efficiently**
  - To allow more concurrency, some special cases do get handled separately.
  - (Search the web for "Escrow Transactions" for example)



- When multiple transactions are executed serially, the database remains in a consistent state. A serializable schedule is equivalent to some serial execution of transactions and thus also maintains the consistency of the database. Therefore, in most DBMSs, serializability is used as the correctness criterion for concurrency control!



## 4.4.7 Lock Protocol



# Locking Protocol

Locking method is the most basic concurrency control method. There maybe many kinds of locking protocols.

## (1) X locks

Only one type of lock, for both read and write.

Compatibility matrix : NL—no lock      X—X lock  
Y —compatible      N—incompatible

|    | NL | X |
|----|----|---|
| NL | Y  | Y |
| X  | Y  | N |

$T_A$   
X\_lock R  
Update R  
⋮  
X\_unlock R  
EOT

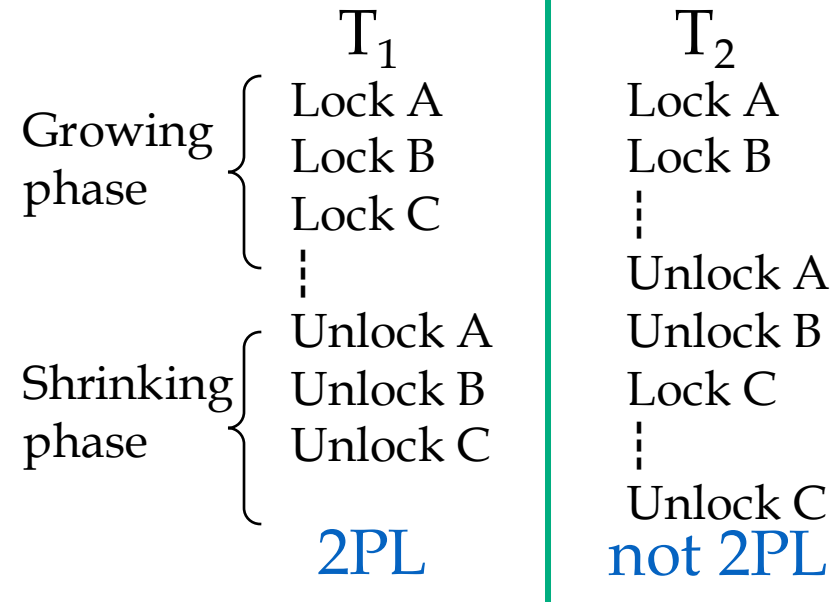
$T_B$   
X\_lock R  
**wait**  
↓  
X\_lock R  
Read R  
⋮



# \*Two Phase Locking

- **Definiton1:** In a transaction, if all locks precede all unlocks, then the transaction is called two phase transaction. This restriction is called ***two phase locking protocol***.
- **Definition2:** In a transaction, if it first acquires a lock on the object before operating on it, it is called ***well-formed***.

- *Theorem:* If S is any schedule of well-formed and two phase transactions, then S is serializable. (proving is omitted)





# Two Phase Locking (2PL)

- **The most common scheme for enforcing conflict serializability**
- **A bit “pessimistic”**
  - Sets locks for fear of conflict... Some cost here.
  - Alternative schemes use multiple versions of data and “optimistically” let transactions move forward
    - Abort when conflicts are detected.
    - Some names to know/look up:
      - Optimistic Concurrency Control
      - Timestamp-Ordered Multi-version Concurrency Control





# Why 2PL guarantees conflict serializability

- When a committing transaction has reached the end of its acquisition phase...
  - Call this the “lock point”
  - At this point, it has *everything it needs* locked...
  - ... and any conflicting transactions either:
    - started release phase before this point
    - are blocked waiting for this transaction
- Visibility of actions of two conflicting transactions are ordered by their lock points
- The order of lock points gives us an equivalent serial schedule!



# Conclusions :

- 1) Well-formed + 2PL : serializable
- 2) Well-formed + 2PL + unlock update at EOT: serializable and recoverable.  
(without domino phenomena)
- 3) Well-formed + 2PL + holding all locks to EOT: strict two phase locking transaction.



# (S,X) locks

(2) (S,X) locks

S lock --- if read access is intended.

X lock --- if update access is intended.

|    | NL | S | X |
|----|----|---|---|
| NL | Y  | Y | Y |
| S  | Y  | Y | N |
| X  | Y  | N | N |

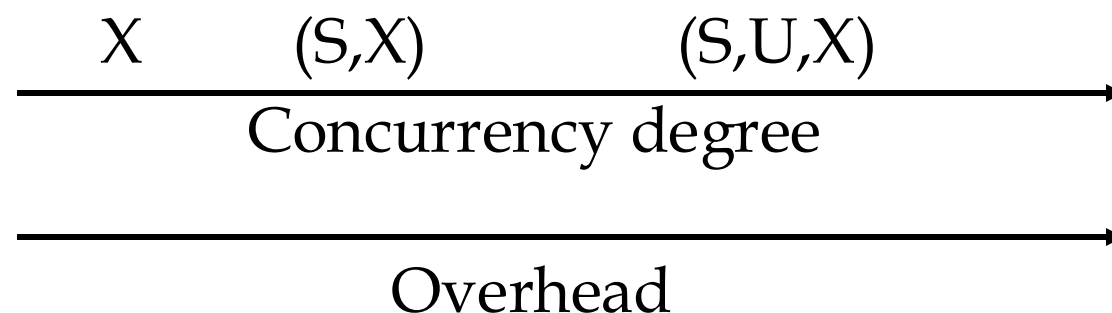


# (S,U,X) locks

U lock --- update lock. For an update access the transaction first acquires a U-lock and then promote it to X-lock.

Purpose: shorten the time of exclusion, so as to boost concurrency degree, and reduce deadlock.

|    | NL | S | U | X |
|----|----|---|---|---|
| NL | Y  | Y | Y | Y |
| S  | Y  | Y | Y | N |
| U  | Y  | Y | N | N |
| X  | Y  | N | N | N |





# Lock Management

- Lock and unlock requests handled by ***Lock Manager***
- LM maintains a ***hash table***, keyed on names of objects being locked.
- LM keeps an entry for each currently held lock
- Entry contains
  - Granted set: Set of xacts currently granted access to the lock
  - Lock mode: Type of lock held (shared or exclusive)
  - Wait Queue: Queue of lock requests

|   | Granted Set | Mode | Wait Queue               |
|---|-------------|------|--------------------------|
| A | {T1, T2}    | S    | T3(X) $\leftarrow$ T4(X) |
| B | {T6}        | X    | T5(X) $\leftarrow$ T7(S) |



# Lock Management (continued)

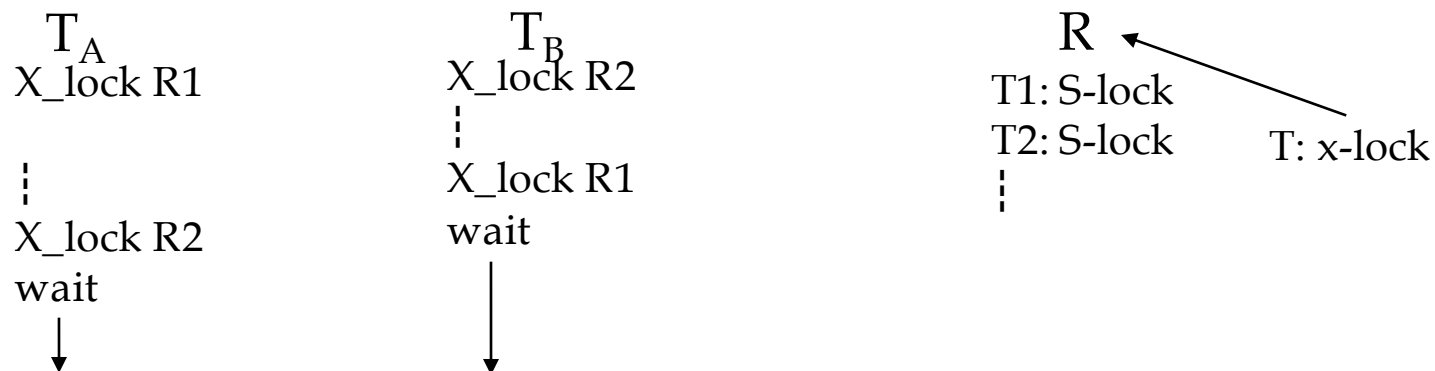
- **When lock request arrives:**
  - Does any xact in Granted Set or Wait Queue want a conflicting lock?
    - If no, put the requester into “granted set” and let them proceed
    - If yes, put requester into wait queue (typically FIFO)
- **Lock upgrade:**
  - Xact with shared lock can request to upgrade to exclusive

|   | Granted Set | Mode | Wait Queue                   |
|---|-------------|------|------------------------------|
| A | {T1, T2}    | S    | <b>T2(X)</b> ← T3(X) ← T4(X) |
| B | {T6}        | X    | T5(X) ← T7(S)                |

# Deadlock & Live Lock

**Dead lock:** wait in cycle, no transaction can obtain all of resources needed to complete.

**Live lock:** although other transactions release their resource in limited time, some transaction can not get the resources needed for a very long time.



- Live lock is simpler, only need to adjust schedule strategy, such as FIFO
- Deadlock: (1) Prevention(don't let it occur); (2) Solving(permit it occurs, but can solve it)



# Deadlocks

- **Deadlock: Cycle of Xacts waiting for locks to be released by each other.**
- **Three ways of dealing with deadlocks:**
  - Prevention
  - Avoidance
  - Detection and Resolution
- **Many systems just punt and use timeouts**
  - What are the dangers with this approach?





# Deadlock Detection

**1. Timeout:** If a transaction waits for some specified time then deadlock is **assumed** and the transaction should be aborted.

**2. Detect deadlock by wait-for graph**  $G=<V,E>$

$V$  : set of transactions  $\{T_i | T_i \text{ is a transaction in DBS } (i=1,2,...n)\}$

$E : \{<T_i, T_j> | T_i \text{ waits for } T_j (i \neq j)\}$

- If there is cycle in the graph, the deadlock occurs.
- When to detect?
  - 1) whenever one transaction waits.
  - 2) periodically



# What to do when detected?

- 1) Pick a victim (youngest, minimum abort cost, ...)
- 2) Abort the victim and release its locks and resources
- 3) Grant a waiter
- 4) Restart the victim (automatically or manually)



# Deadlock avoidance

- 1) Requesting all locks at initial time of transaction.
- 2) Requesting locks in a specified order of resource.
- 3) Abort once conflicted.
- 4) Transaction Retry



# Deadlock avoidance

- Every transaction is uniquely time stamped. If  $T_A$  requires a lock on a data object that is already locked by  $T_B$ , one of the following methods is used:
  - a) Wait-die:**  $T_A$  waits if it is older than  $T_B$ , otherwise it “dies”, i.e. it is aborted and automatically retried with original timestamp.
  - b) Wound-wait:**  $T_A$  waits if it is younger than  $T_B$ , otherwise it “wound”  $T_B$ , i.e.  $T_B$  is aborted and automatically retried with original timestamp.

*In above, both have only one direction wait, either older  $\rightarrow$  younger or younger  $\rightarrow$  older. It is impossible to occur wait in cycle, so the dead lock is avoided.*



# Lock Granularity



# Lock Granularity

- Hard to decide what granularity to lock
  - Tuples vs Pages vs Tables?
- What is the tradeoff?
  - Fine-grained availability of resources would be nice (e.g. lock per tuple)
  - Small # of locks to manage would also be nice (e.g. lock per table)
  - Can't have both!
    - Or can we???

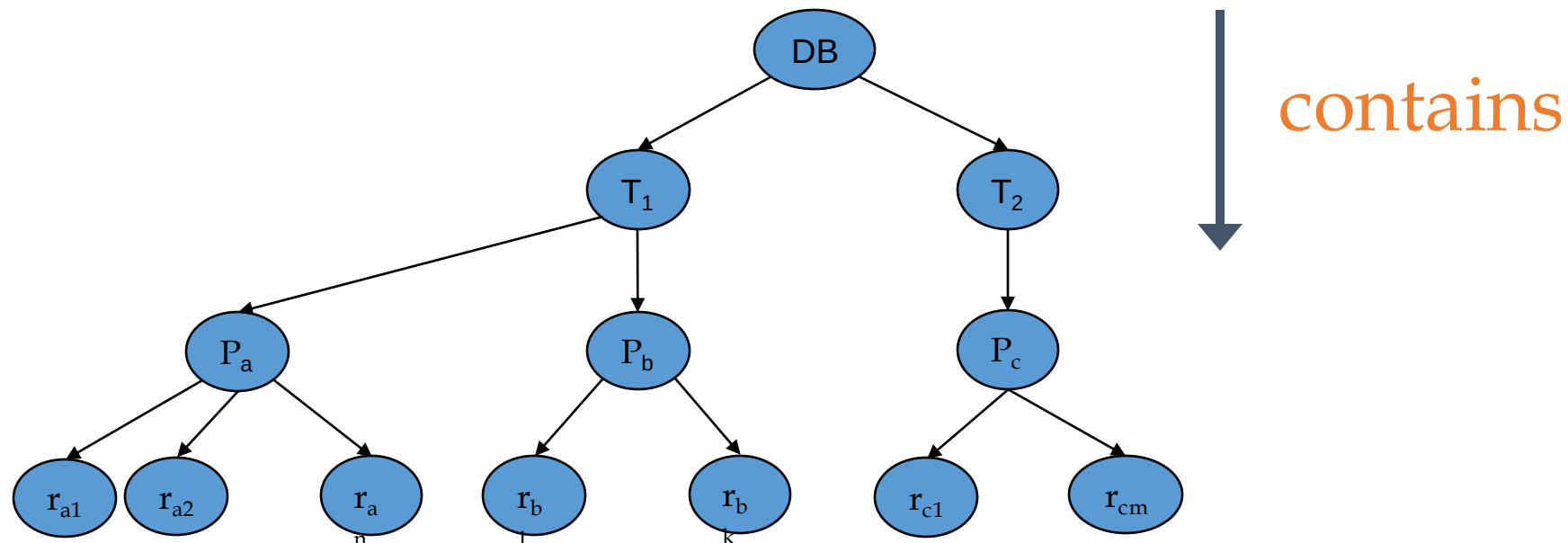


# Multiple Locking Granularity

- **Shouldn't have to make same decision for all transactions!**
- Allow data items to be of various sizes
- Define a hierarchy of data granularities, small nested within large
  - Can be represented graphically as a tree.

# Example of Granularity Hierarchy (RDBMS)

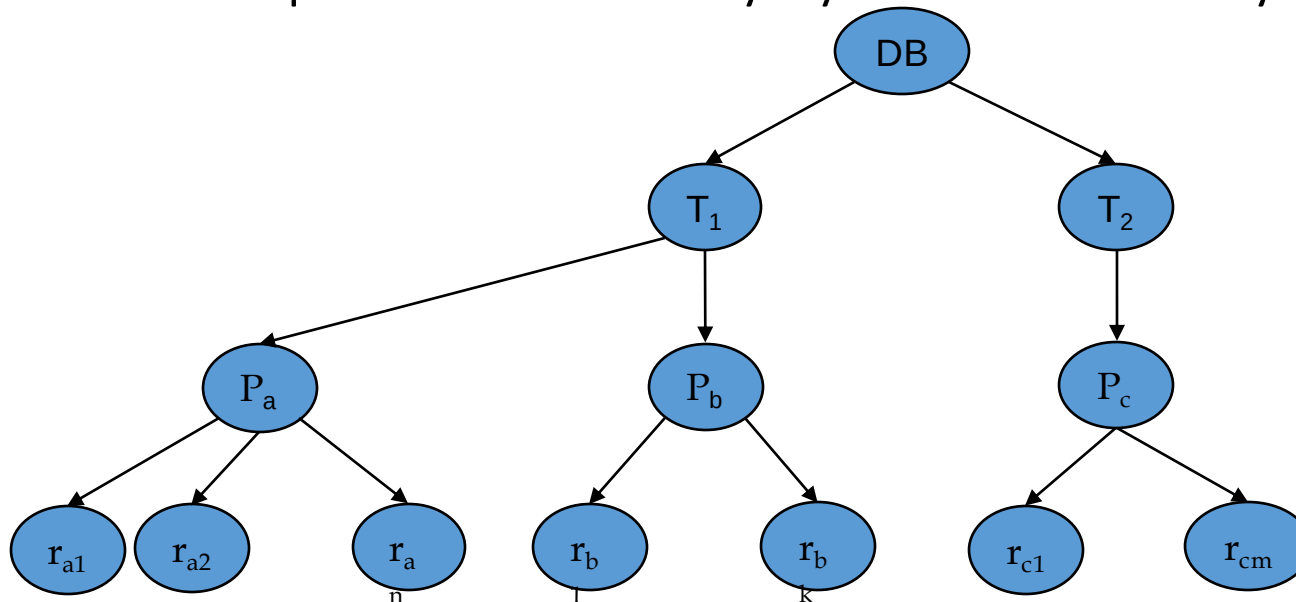
- Data “containers” can be viewed as nested.
- The levels, starting from the coarsest (top) level are
  - Database, Tables, Pages, Records
- When a transaction locks a node in the tree *explicitly*, it *implicitly* locks all the node’s descendants in the same mode.





# Multiple Locking Granularity

- Granularity of locking (level in tree where locking is done):
  - **Fine granularity** (lower in tree): High concurrency, lots of locks (overhead)
  - **Coarse granularity** (higher in tree): Few locks (low overhead), lost concurrency
    - Lost potential concurrency if you don't need everything inside the coarse grain





# Real-World Locking Granularities

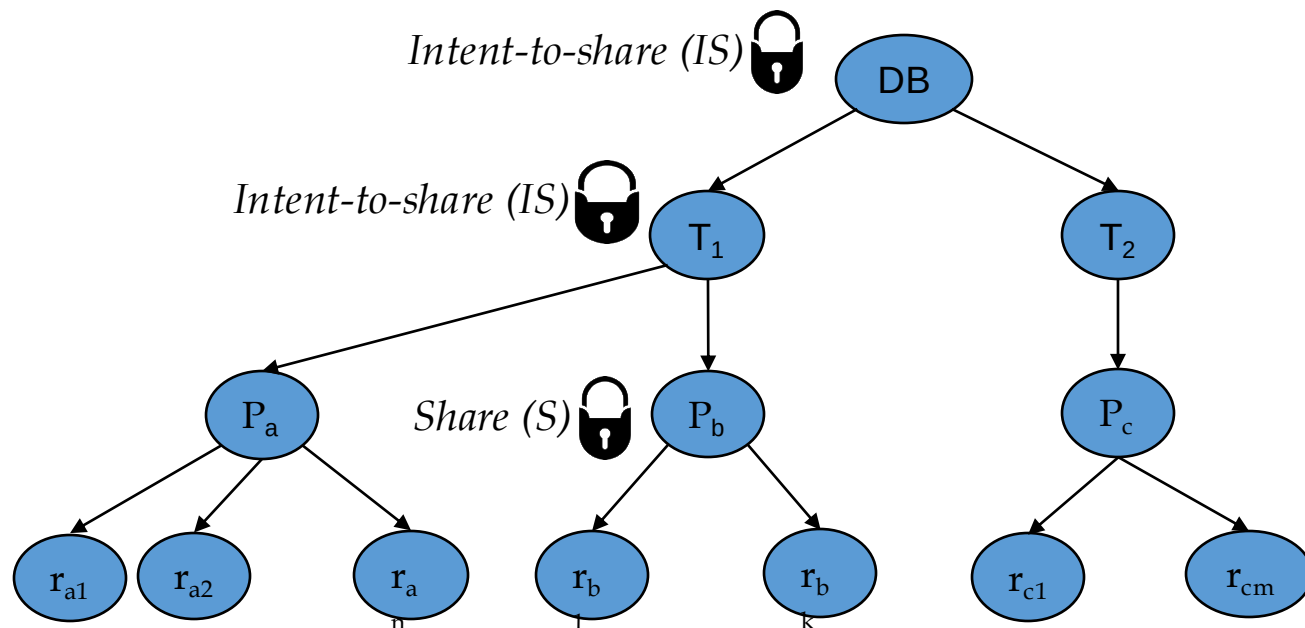
| Resource        | Description                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| RID             | A row identifier used to lock a single row within a heap.                                                                    |
| KEY             | A row lock within an index used to protect key ranges in serializable transactions.                                          |
| PAGE            | An 8-kilobyte (KB) page in a database, such as data or index pages.                                                          |
| EXTENT          | A contiguous group of eight pages, such as data or index pages.                                                              |
| HoBT            | A heap or B-tree. A lock protecting a B-tree (index) or the heap data pages in a table that does not have a clustered index. |
| TABLE           | The entire table, including all data and indexes.                                                                            |
| FILE            | A database file.                                                                                                             |
| APPLICATION     | An application-specified resource.                                                                                           |
| METADATA        | Metadata locks.                                                                                                              |
| ALLOCATION_UNIT | An allocation unit.                                                                                                          |
| DATABASE        | The entire database.                                                                                                         |

From MS SQL Server

[https://technet.microsoft.com/en-us/library/jj856598\(v=sql.110\).aspx](https://technet.microsoft.com/en-us/library/jj856598(v=sql.110).aspx)

# Solution: New Lock Modes, Protocol

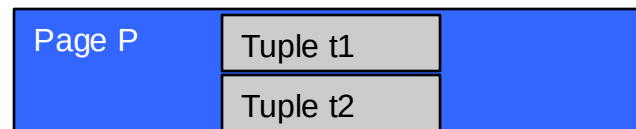
- Allow transactions to lock at each level, but with a special protocol using new “intent” locks:
- Before getting S or X lock, Xact must have proper intent locks on all its ancestors in the granularity hierarchy.





# New Lock Modes – Intention Lock Modes

- 3 additional lock modes:
  - **IS**: Intent to get S lock(s) at finer granularity.
  - **IX**: Intent to get X lock(s) at finer granularity.
  - **SIX**: Like S & IX at the same time. Why useful?
- Intention locks allow a higher level node to be locked in S or X mode without having to check all descendent nodes





# Multiple Granularity Locking Protocol

- Each Xact starts from the root of the hierarchy.
  - To get S or IS lock on a node, must hold IS or IX on parent node.
    - What if Xact holds S on parent? SIX on parent?
  - To get X or IX or SIX on a node, must hold IX or SIX on parent node.
  - Must release locks in bottom-up order.
- 
- 2-phase and lock compatibility matrix rules enforced as well
  - Protocol is correct in that it is ***equivalent to directly setting locks at leaf levels of the hierarchy.***

Database  
|  
Tables  
|  
Pages  
|  
Tuples



# Lock Compatibility Matrix

- IS – Intent to get S lock(s) at finer granularity.
- IX – Intent to get X lock(s) at finer granularity.
- SIX mode: Like S & IX at the same time.

Handy simple case to remember:  
Could 2 intent locks be compatible?

|        |          |   |    |
|--------|----------|---|----|
| Page P | Tuple t1 | S | IS |
|        | Tuple t2 | X | IX |

|     | IS | IX | S     | SIX | X     |
|-----|----|----|-------|-----|-------|
| IS  |    |    |       |     |       |
| IX  |    |    |       |     |       |
| S   |    |    | true  |     | false |
| SIX |    |    |       |     |       |
| X   |    |    | false |     | false |

Database  
|  
Tables  
|  
Pages  
|  
Tuples



# Lock Compatibility Matrix, Cont

- IS – Intent to get S lock(s) at finer granularity.
- IX – Intent to get X lock(s) at finer granularity.
- SIX mode: Like S & IX at the same time.

Handy simple case to remember:  
Could 2 intent locks be compatible?

|        |          |   |    |
|--------|----------|---|----|
| Page P | Tuple t1 | S | IS |
|        | Tuple t2 | X | IX |

|     | IS    | IX    | S     | SIX   | X     |
|-----|-------|-------|-------|-------|-------|
| IS  | true  | true  | true  | true  | false |
| IX  | true  | true  | false | false | false |
| S   | true  | false | true  | false | false |
| SIX | true  | false | false | false | false |
| X   | false | false | false | false | false |

Database  
|  
Tables  
|  
Pages  
|  
Tuples



# Real-World Lock Compatibility Matrix

|       | NL | SCH-S | SCH-M | S | U | X | IS | IU | IX | SIU | SIX | UIX | BU | RS-S | RS-U | RI-N | RI-S | RI-U | RI-X | RX-S | RX-U | RX-X |
|-------|----|-------|-------|---|---|---|----|----|----|-----|-----|-----|----|------|------|------|------|------|------|------|------|------|
| NL    | N  | N     | N     | N | N | N | N  | N  | N  | N   | N   | N   | N  | N    | N    | N    | N    | N    | N    | N    | N    | N    |
| SCH-S | N  | N     | C     | N | N | N | N  | N  | N  | N   | N   | N   | N  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| SCH-M | N  | C     | C     | C | C | C | C  | C  | C  | C   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| S     | N  | N     | C     | N | N | C | N  | N  | C  | N   | C   | C   | C  | N    | N    | N    | N    | N    | C    | N    | N    | C    |
| U     | N  | N     | C     | N | C | C | N  | C  | C  | C   | C   | C   | C  | N    | C    | N    | N    | C    | C    | N    | C    | C    |
| X     | N  | N     | C     | C | C | C | C  | C  | C  | C   | C   | C   | C  | C    | C    | N    | C    | C    | C    | C    | C    | C    |
| IS    | N  | N     | C     | N | N | C | N  | N  | N  | N   | N   | N   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| IU    | N  | N     | C     | N | C | C | N  | N  | N  | N   | N   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| IX    | N  | N     | C     | C | C | C | N  | N  | N  | C   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| SIU   | N  | N     | C     | N | C | C | N  | N  | C  | N   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| SIX   | N  | N     | C     | C | C | C | N  | N  | C  | C   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| UIX   | N  | N     | C     | C | C | C | N  | C  | C  | C   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| BU    | N  | N     | C     | C | C | C | C  | C  | C  | C   | C   | C   | C  | I    | I    | I    | I    | I    | I    | I    | I    | I    |
| RS-S  | N  | I     | I     | N | N | C | I  | I  | I  | I   | I   | I   | I  | N    | N    | C    | C    | C    | C    | C    | C    | C    |
| RS-U  | N  | I     | I     | N | C | C | I  | I  | I  | I   | I   | I   | I  | N    | C    | C    | C    | C    | C    | C    | C    | C    |
| RI-N  | N  | I     | I     | N | N | N | I  | I  | I  | I   | I   | I   | I  | C    | C    | N    | N    | N    | N    | C    | C    | C    |
| RI-S  | N  | I     | I     | N | N | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | N    | N    | N    | C    | C    | C    | C    |
| RI-U  | N  | I     | I     | N | C | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | N    | N    | C    | C    | C    | C    | C    |
| RI-X  | N  | I     | I     | C | C | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | N    | C    | C    | C    | C    | C    | C    |
| RX-S  | N  | I     | I     | N | N | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | C    | C    | C    | C    | C    | C    | C    |
| RX-U  | N  | I     | I     | N | C | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | C    | C    | C    | C    | C    | C    | C    |
| RX-X  | N  | I     | I     | C | C | C | I  | I  | I  | I   | I   | I   | I  | C    | C    | C    | C    | C    | C    | C    | C    | C    |

## Key

|       |                           |      |                              |
|-------|---------------------------|------|------------------------------|
| N     | No Conflict               | SIU  | Share with Intent Update     |
| I     | Illegal                   | SIX  | Shared with Intent Exclusive |
| C     | Conflict                  | UIX  | Update with Intent Exclusive |
|       |                           | BU   | Bulk Update                  |
| NL    | No Lock                   | RS-S | Shared Range-Shared          |
| SCH-S | Schema Stability Locks    | RS-U | Shared Range-Update          |
| SCH-M | Schema Modification Locks | RI-N | Insert Range-Null            |
| S     | Shared                    | RI-S | Insert Range-Shared          |
| U     | Update                    | RI-U | Insert Range-Update          |
| X     | Exclusive                 | RI-X | Insert Range-Exclusive       |
| IS    | Intent Shared             | RX-S | Exclusive Range-Shared       |
| IU    | Intent Update             | RX-U | Exclusive Range-Update       |
| IX    | Intent Exclusive          | RX-X | Exclusive Range-Exclusive    |

From MS SQL Server

[https://technet.microsoft.com/en-us/library/jj856598\(v=sql.110\).aspx](https://technet.microsoft.com/en-us/library/jj856598(v=sql.110).aspx)







# Summary

- **Correctness criterion for isolation is “serializability”.**
  - In practice, we use “conflict serializability” which is conservative but easy to enforce
- **Two Phase Locking and Strict 2PL: Locks implement the notions of conflict directly**
  - The lock manager keeps track of the locks issued.
  - **Deadlocks** may arise; can either be prevented or detected.
- **Multi-Granularity Locking:**
  - Allows flexible tradeoff between lock “scope” in DB, and # of lock entries in lock table
- **More to the story**
  - Optimistic/Multi-version/Timestamp CC
  - Index “latching”, phantoms
  - Actually, there’s much much more :-)